3. FIFO CPU Scheduling

Code:

```c
#include <stdio.h>

int main()
{
    int A[100][4];
    int i, j, n, total = 0, index, temp;
    float avg_wt, avg_tat;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter Burst Time:\n");

    for (i = 0; i < n; i++)
    {
        printf("P%d: ", i + 1);
        scanf("%d", &A[i][1]);
        A[i][0] = i + 1;
    }

    // Sort processes according to Burst Time
    for (i = 0; i < n; i++)
    {
        index = i;

        for (j = i + 1; j < n; j++)
        {
            if (A[j][1] < A[index][1])
                index = j;
        }

        temp = A[i][1];
```

```
        A[index][1] = temp;

        temp = A[i][0];
        A[i][0] = A[index][0];
        A[index][0] = temp;
    }

    // Calculate Waiting Time
    A[0][2] = 0;

    for (i = 1; i < n; i++)
    {
        A[i][2] = 0;

        for (j = 0; j < i; j++)
            A[i][2] += A[j][1];

        total += A[i][2];
    }

    avg_wt = (float)total / n;

    // Calculate Turnaround Time
    total = 0;

    printf("\nP\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++)
    {
        A[i][3] = A[i][1] + A[i][2];
        total += A[i][3];
```

```
for (i = 0; i < n; i++)
{
    A[i][3] = A[i][1] + A[i][2];
    total += A[i][3];

    printf("P%d\t%d\t%d\t%d\n",
           A[i][0], A[i][1], A[i][2], A[i][3]);
}

avg_tat = (float)total / n;

printf("\nAverage Waiting Time = %.2f", avg_wt);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat);

return 0;
```

Output:

```
Enter number of processes: 3
Enter Burst Time:
P1: 2
P2: 5
P3: 3

P       BT      WT      TAT
P1      2       0       2
P3      3       2       5
P2      5       5       10

Average Waiting Time = 2.33
Average Turnaround Time = 5.67

--------------------------------
Process exited after 7.749 seconds with return value 0
Press any key to continue . . .
```